

# CSA1280 – COMPUTER ARCHITECTURE

## EXPERIMENT – 5

### MULTIPLICATION OF TWO 8-BIT NUMBERS

**AIM:** To write and execute an 8085 assembly language program to Multiply two 8-bit numbers and store the result in memory.

#### ALGORITHM

1. Load the first number into register **B**.
2. Load the second number into register **C**.
3. Clear register pair **HL** to store the result.
4. Add the first number repeatedly using the second number as the counter.
5. Decrement the counter after each addition.
6. Repeat until the counter becomes zero.
7. The 16-bit multiplication result is stored in **HL**.
8. Stop the program.

#### CODE:

```
MVI B,05H  
MVI C,04H  
MVI A,00H  
LOOP: ADD B  
DCR C  
JNZ LOOP  
STA 8050H  
HLT
```

## OUTPUT:

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

| Register | Value |
|----------|-------|
| A        | 0F    |
| BC       | 05 00 |
| DE       | 00 00 |
| HL       | 00 00 |
| PSW      | 00 00 |
| PC       | 42 13 |
| SP       | FF FF |
| Int-Reg  | 00    |

Flag

| Flag | Value |
|------|-------|
| S    | 0     |
| Z    | 1     |
| AC   | 0     |
| P    | 1     |
| C    | 0     |

Decimal - Hex Conversion

| Decimal | Hex |
|---------|-----|
| 0       | 0   |

I/O Ports

| Port | Value |
|------|-------|
| 0    | 00    |

Memory

| Address | Value |
|---------|-------|
| 2501    | 03    |

Load me at

```
1 LDA 2500
2 MOV B,A
3
4 LDA 2501
5 MOV C,A
6
7 MVI A,00
8
9 LOOP: ADD B
10 DCR C
11 JNZ LOOP
12
13 STA 2502
14
15 HLT
16
```

Memory

| Address (Hex) | Address | Data |
|---------------|---------|------|
| 09C4          | 2500    | 5    |
| 09C5          | 2501    | 3    |
| 09C6          | 2502    | 15   |
| 09C7          | 2503    | 0    |
| 09C8          | 2504    | 0    |
| 09C9          | 2505    | 0    |
| 09CA          | 2506    | 0    |
| 09CB          | 2507    | 0    |
| 09CC          | 2508    | 0    |
| 09CD          | 2509    | 0    |
| 09CE          | 2510    | 0    |
| 09CF          | 2511    | 0    |
| 09D0          | 2512    | 0    |
| 09D1          | 2513    | 0    |

Line No Assembler Message

```
0 Program assembled successfully
```

Simulator: Idle

Windows Taskbar: Search, 14:00, 25-07-2026